

8 (a) (i) 1.
$$d = \frac{\text{velocity of droplets}}{\text{frequency of droplet emission}}$$

$$= \frac{20}{110000}$$

$$= 1.8182 \times 10^{-4} \text{ m} = 182 \mu\text{m}$$

2. mass of droplet $= \rho \left(\frac{4}{3} \pi r^3 \right)$

$$= \left(\frac{0.84 \times 10^{-3}}{10^{-6}} \right) \left(\frac{4}{3} \pi \right) \left(\frac{80 \times 10^{-6}}{2} \right)^3$$

$$= 2.2519 \times 10^{-10} = 2.25 \times 10^{-10} \text{ kg}$$

(ii) 1.
$$F = \frac{q^2}{4\pi\epsilon_0 d^2}$$

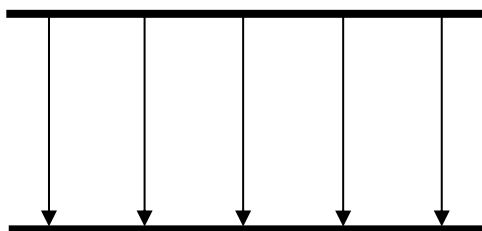
$$= \frac{(-1.0 \times 10^{-13})^2}{4\pi (8.85 \times 10^{-12}) (182 \times 10^{-6})^2}$$

$$= 2.71 \times 10^{-9} \text{ N} = 2.7 \text{ nN (shown)}$$

$$\begin{aligned}
 2. \quad a &= \frac{F}{m} \\
 &= \frac{2.71 \times 10^{-9}}{2.25 \times 10^{-10}} \\
 &= 12.0444 = 12.0 \text{ m s}^{-2}
 \end{aligned}$$

- (iii) The “guard droplets” increase the distance between the charged droplets, reducing the large accelerations due to the electrostatic forces, which could have caused uneven or irregular printing due to uneven spacing of the droplets / changing velocities of droplets / droplets colliding with one another.

(b) (i)



(ii) $E = \frac{\Delta V}{d} = \frac{10000}{0.010} = 1.0 \times 10^6 \text{ N C}^{-1}$

- (iii) 1. Perpendicularly out of the page.

2. The magnetic force provides the centripetal force.

$$\begin{aligned}
 Bqv &= \frac{mv^2}{r} \\
 B &= \frac{mv}{rq} \\
 &= \frac{(2.25 \times 10^{-10})(20)}{(0.260)(1.0 \times 10^{-13})} \\
 &= 1.7308 \times 10^5 = 1.73 \times 10^5 \text{ T}
 \end{aligned}$$

It is not practical / feasible to use a magnetic field to deflect an ink droplet as the required magnetic flux density is too large. (As a reference, MRI machines operate at around 1.5 T to 3 T, and those are already considered very strong magnetic fields that require many safety protocols during its operation!)

(c) (i) $Q = Pt$
 $= I^2 Rt$
 $= (0.50)^2 (30)(0.010 \times 10^{-6})$
 $= 7.5 \times 10^{-8} \text{ J}$

(ii) $Q = mc\Delta\theta + ml_v = 7.5 \times 10^{-8} \text{ J}$
 $m = \frac{7.5 \times 10^{-8}}{(2090)(80 - 25) + (444 \times 10^3)}$
 $= 1.3418 \times 10^{-13} = 1.34 \times 10^{-13} \text{ kg}$

Raffles Institution
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(iii)
$$V = \frac{m}{\rho} = \frac{1.34 \times 10^{-13}}{(1.17 \times 10^{-3})/10^{-6}} = 1.1453 \times 10^{-16} \text{ m}^3$$
$$x = \frac{1.1453 \times 10^{-16}}{(20 \times 10^{-6})^2}$$
$$= 2.8632 \times 10^{-7}$$
$$= 0.286 \times 10^{-6} \text{ m} = 0.286 \mu\text{m}$$

(iv) There are significant heat losses to the surroundings and not all of the heat produced from the heating element goes into heating the ink layer.

(d) Advantage: Thermal inkjet printing has much higher resolution, resulting in better image quality, due to smaller droplet sizes for thermal inkjet printer at 10 μm compared to CIJ printer's at 80 μm .

Disadvantage: Thermal inkjet printing has much smaller printing speeds due to a lower frequency of droplet ejection for thermal inkjet printer at 18 kHz compared to CIJ printer's at 110 kHz.